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MIL-E-0022200/2B(SHIPS)
AMENDMENT-2
15 September 1970
SUPERSEDING 1/
AMENDMENT-1
6 June 1969

MILITARY SPECIFICATION

ELECTRODES, WELDING, COVERED (AUSTENITIC CHROMIUM-NICKEL STEEL, FOR CORROSIVE AND HIGH TEMPERATURE SERVICES)

This amendment forms a part of Military Specification MIL-E-0022200/2B(SHIPS), 3 January 1969.

Pages 1 and 2

Table I: Delete and substitute:

"Table I - Classification.

Type	Class	Positions of welding	Sizes (diameter) inch
MIL-308-15 MIL-308-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-308Co-15 MIL-308Co-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-308L-15 MIL-308L-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32 1/8, 5/32 3/16, 1/4
MIL-308CoL-15	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-308HC-15 MIL-308HC-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-309-15 MIL-309-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32 1/8, 5/32 3/16, 1/4
MIL-309Co-15 MIL-309Co-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4

1/ CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

FSC 3439

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Table I - Classification (cont'd).

Type	Class	Positions of welding	Sizes (diameter) inch
MIL-309L-15 MIL-309L-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-309CoL-15 MIL-309CoL-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-309Cb-15 MIL-309Cb-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-310-15 MIL-310-16	1 2 3	All Horizontal fillets and flat Flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4 5/16
MIL-312-15 MIL-312-16	1 2 3	All Horizontal fillets and flat Flat	5/64, 3/32, 1/8, 5/32 3/16, 1/4 5/16
MIL-316-15 MIL-316-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-316L-15 MIL-316L-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-317-15 MIL-317-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-318-15 MIL-318-16	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 1/4
MIL-330-15 MIL-330-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-347-15 MIL-347-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-347Co-15 MIL-347Co-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-347HC-15 MIL-347HC-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-349-15 MIL-349-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4
MIL-16.8.2-15 MIL-16.8.2-16	1 2	All Horizontal fillets and flat	1/16, 5/64, 3/32, 1/8, 5/32 3/16, 1/4

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2.2: Add:

"AMERICAN WELDING SOCIETY (AWS)
USA Standard Z49.1 - Safety in Welding and Cutting.

(Application for copies should be addressed to American Welding Society, Inc., 345 East 47th Street, New YOrk, N.Y. 10017.)"

Add 3.2.2:

"3.2.2 Extent of covering at arc ends. Tubular electrodes at the arc end may extend beyond the coating in excess of the 1/32-inch requirement of 3.5.9.1, MIL-E-0022200 provided the actual volume of the exposed electrode is not greater than

$$1/32\,\pi\left(\frac{\text{Nominal diameter}}{2}\right)^2 \text{"}.$$

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Table III: Delete and substitute:

"Table III - Chemical composition (percent) of deposited weld metal 1/.

Type	Percentage (single values are maximum)										
	Carbon	Manganese	Phosphorus	Sulfur	Silicon	Chromium	Nickel	Molybdenum	Columbium	Tungsten	Cobalt
2/ MIL-308-15 and 16	0.08	2.5	0.04	0.03	0.90	18.0-21.0	9.0-11.0	---	---	---	---
2/ MIL-308Co-15 and 16	.08	2.5	.04	.03	.90	18.0-21.0	9.0-11.0	---	---	---	0.10
2/ MIL-308L-15 and 16	.04	2.5	.04	.03	.90	18.0-21.0	9.0-11.0	---	---	---	---
2/ MIL-308CoL-15	.04	2.5	.04	.03	.90	18.0-21.0	9.0-11.0	---	---	---	0.10
3/ MIL-308HC 15 and 16	.08-.15	2.5	.03	.03	4/	18.0-21.0	9.0-11.0	---	---	---	---
2/ MIL-309-15 and 16	.10	2.5	.04	.03	0.90	22.0-25.0	12.0-14.0	---	---	---	---
2/ MIL-309Co-15 and 16	.10	2.5	.04	.03	.90	22.0-25.0	12.0-14.0	---	---	---	0.10
2/ MIL-309L-15 and 16	.04	2.5	.04	.03	.90	22.0-25.0	12.0-14.0	---	---	---	---
2/ MIL-309CoL-15 and 16	.04	2.5	.04	.03	.90	22.0-25.0	12.0-14.0	---	---	---	0.10
2/ MIL-309Cb-15 and 16	.12	2.5	.04	.03	.90	22.0-25.0	12.0-14.0	--	Cb&Ta 0.70-1.00	---	---
MIL-310-15 and 16	.20	2.5	.03	.03	.75	25.0-28.0	20.0-22.5	---	---	---	---
MIL-312-15 and 16	.15	2.5	.04	.03	.90	28.0-32.0	8.0-10.5	---	---	---	---
MIL-316-15 and 16	.08	2.5	.04	.03	.90	17.0-20.0	11.0-14.0	2.0-2.5	---	---	---
MIL-316L-15 and 16	.04	2.5	.04	.03	.90	17.0-20.0	11.0-14.0	2.0-2.5	---	---	---
MIL-317-15 and 16	.08	2.5	.04	.03	.90	18.0-21.0	12.0-14.0	3.0-4.0	---	---	---
MIL-318-15 and 16	.08	2.5	.04	.03	.90	17.0-20.0	11.0-14.0	2.0-2.5	5/	---	---
MIL-330-15 and 16	.25	2.5	.04	.03	.90	14.0-17.0	33.0-37.0	---	---	---	---
2/ MIL-347-15 and 16	.08	2.5	.04	.03	.90	18.0-21.0	9.0-11.0	---	6/	---	---
2/ MIL-347Co-15 and 16	.08	2.5	.04	.03	.90	18.0-21.0	9.0-11.0	---	6/	---	0.10
MIL-347HC-15 and 16	.08-.15	2.5	.025	.025	4/	18.0-21.0	9.0-11.0	---	7/	---	---
MIL-349-15 and 16	.07-.13	2.5	.03	.03	0.90	18.0-21.0	8.0-10.0	0.35-0.65	8/ 0.75-1.2	1.25-1.75	---
MIL-16.8.2-15 and 16	.10	2.5	.03	.03	.50	14.5-16.5	7.5-9.5	1.0-2.0	---	---	---

See footnotes at top of next page.

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Page 5 (cont'd)

1/ Analysis shall be made for the elements for which specific values are shown in the table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that the total of these other elements, except iron, is not present in excess of 0.70 percent.
2/ Delta ferrite content shall be from 4 to 9 percent determined by Magne-Gage. (See figure 4 and notes.)
3/ Delta ferrite content shall not exceed 4 percent determined by Magne-Gage. (See figure 4 and notes.)
4/ Silicon shall be not more than 5 times actual carbon present.
5/ Columbium (combined Cb+Ta) shall be not less than 6 times actual carbon present but not more than 1.00 percent.
6/ Columbium (combined Cb+Ta) shall be not less than 8 times actual carbon present but not more than 1.00 percent. Tantalum shall not exceed 0.10 percent.
7/ Columbium (combined Cb+Ta) shall be not less than 8 times actual carbon present but not more than 1.4 percent.
8/ Titanium shall be limited to 0.15 percent maximum."

Page 6

Table VI: Delete and substitute:

"Table VI - Summary of tests required for quality conformance inspection.

Type	Required tests			Procedures and requirements
	Chemical analysis	Ferrite content test No. 4	Groove weld usability test No. 8a	
MIL-308	X	X	X	For chemistry use 4.6.8 of MIL-E-0022200E and table III herein
MIL-308Co	X	X	X	
MIL-308L	X	X	X	
MIL-308CoL	X	X	X	
MIL-308HC	X	X	X	
MIL-309	X	X	X	For ferrite content, see table VII and notes thereto
MIL-309Co	X	X	X	
MIL-309L	X	X	X	
MIL-309CoL	X	X	X	
MIL-309Cb	X	X	X	
MIL-310	X	-	X	
MIL-312	X	-	X	
MIL-316	X	-	X	
MIL-316L	X	-	X	
MIL-317	X	-	X	
MIL-318	X	-	X	For weld test 8a, see table VII
MIL-330	X	-	X	
MIL-347	X	X	X	
MIL-347Co	X	X	X	
MIL-347HC	X	-	X	
MIL-349	X	-	X	
MIL-16.8.2	X	-	X	
All	Diameter and length of core wire Concentricity			Table VIII

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Table VII: Delete and substitute:

"Table VII - Summary of required weld tests[1/]

Weld test No.	Electrode Type	Diameter Sizes 1/16	5/64	3/32	1/8	5/32	3/16	1/4	5/16	Weld Joint Design Procedure	Welding Position[2/]	Tests: X-ray	Tensile	Bend	Chemical analysis[3/]	Ferrite content[4/]	Requirements
3	All Types	-	-	X	X	X	X	X	X	Figure 1 and Notes	F	X	X	X	X	-	3.7.1 of MIL-E-0022200E and Tables II and III herein. 4.6.8 of MIL-E-0022200E
6	All Types	X	X	-	-	-	-	-	-	Figure 2 and Notes	F	X	X	-	X	-	3.7.1 of MIL-E-0022200E and Tables II and III herein. 4.6.8 of MIL-E-0022200E
8a	All Types	X	X	X	X	X	-	-	-	Figure 3 and Notes	V	X	-	-	-	-	3.7.1 of MIL-E-0022200E
		-	-	-	-	-	X	X	X		F	X[5/]	-	-	-	-	
4	MIL-308 MIL-308Co MIL-308L MIL-308CoL MIL-308HC MIL-309 MIL-309Co MIL-309L MIL-309CoL MIL-309Cb MIL-347 MIL-347Co	X	X	X	X	X	X	X	X	Figure 4 and Notes	F	-	-	-	-	X	Table III herein

1/ For qualification testing of the electrodes for the Qualified Products List (QPL), all electrode types shall be tested using direct current reverse polarity. In addition, MIL-XXX-16 electrode types shall be tested using alternating current. For comparison inspection and quality conformance inspection, all electrode types shall be tested using direct current reverse polarity.

2/ F - flat V - vertical

3/ Chemical analysis shall be obtained by the procedures and methods described in 4.6.8 of MIL-E-0022200E, except the ferrite test pad may be used in lieu of the weld pad required in 4.6.8 (MIL-E-0022200E) for obtaining chips for those electrode types requiring ferrite determination.

4/ Delta ferrite content of weld metal shall be determined by the Magne-Gage calibrated against specimens having known delta ferrite contents.

5/ For qualification and comparison inspection testing, the 8a test may be eliminated for 3/16, 1/4, and 5/16-inch diameter sizes."

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5.3, 5.3.1 and 5.3.2: Delete and substitute:

"5.3 Marking.

"5.3.1 Interior packages. Shipment marking information shall be provided on interior packages in accordance with the manufacturer's commercial practice. In addition to the lot identification required by 4.4.6 of MIL-E-0022200E, the information shall include type, size, specification number, manufacturer's or distributor's name, date of manufacture (month/year), and manufacturer's or distributor's brand or type designation. In addition, all packages, or the smallest integral unit within a shipping container, shall carry the following caution label, as a minimum, prominently displayed in legible type on the package:

CAUTION
Welding may produce fumes and gases hazardous to health.
Avoid breathing these fumes and gases.
Use adequate ventilation.
See USA Standard Z49.1 'Safety in Welding and Cutting'
published by the American Welding Society.

Significant toxic constituents when present in the electrode in greater than trace amounts shall be identified on the warning label so that normal ventilation can be increased accordingly. These constituents include but are not limited to those itemized in section 8.5 through 8.11 of USA Z49.1.

"5.3.2 Exterior shipping containers and palletized unit loads. Shipment marking information shall be provided on exterior shipping containers and palletized unit loads in accordance with the manufacturer's commercial practice. In addition to the information required for interior packages as specified in 5.3.1 except the caution label, the information shall include shipping destination, stock number furnished with contract or order, customer's order number, customer's item number, and customer's name."

Pages 9 and 10

6.1 through 6.1.17: Delete and substitute:

"6.1 Intended use.

"6.1.1 MIL-308-15 and 16. These electrodes are intended for the general welding of 18 percent chromium, 8 percent nickel, corrosion-resistant (stainless) steels.

"6.1.2 MIL-308Co-15 and 16. These electrodes are intended for the welding applications of 6.1.1 requiring a limited cobalt content.

"6.1.3 MIL-308L-15 and 16. These electrodes are intended for the general welding of 18 percent chromium, 8 percent nickel extra low carbon corrosion-resistant steel, used where intergranular corrosion must be limited but where corrosive conditions are not the most severe.

"6.1.4 MIL-308CoL-15. These electrodes are intended for the welding applications of 6.1.3 requiring a limited cobalt content.

"6.1.5 MIL-308HC-15 and 16. These electrodes are intended for applications where low magnetic permeability is of prime importance and where corrosion is not a factor.

"6.1.6 MIL-309-15 and 16. These electrodes are intended for the joining of the 18-8 grades of corrosion-resistant steel to mild steel; special treatment steel, class B armor, and other low alloy air hardening steels, when specified; and carbon and straight chromium steels to austenitic grades. These electrodes are also for deposition of a corrosion-resistant overlay on carbon and low alloy steels.

"6.1.7 MIL-309Co-15 and 16. These electrodes are intended for the welding applications of 6.1.6 requiring a limited cobalt content.

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Pages 9 and 10 (cont'd)

"6.1.8 MIL-309L-15 and 16. These electrodes are intended for similar applications as MIL-309-15 and 16 types, but where added resistance to intergranular corrosion is desired.

"6.1.9 MIL-309CoL-15 and 16. These electrodes are intended for the welding applications of 6.1.8 requiring a limited cobalt content.

"6.1.10 MIL-309CB-15 and 16. These electrodes are intended for similar applications as the MIL-309-15 and 16 types, but where expected corrosive conditions are more severe, or higher strength at elevated temperatures is desirable.

"6.1.11 MIL-310-15 and 16. These electrodes are intended for the welding of 25 percent chromium, 20 percent nickel heat-resistant alloy steels used in high temperature service; special treatment steel, class B armor, and other low-alloy air hardening steels, where specified; and carbon and straight chromium steels to austenitic grades. These electrodes are also intended for deposition of a corrosion-resistant overlay on carbon and low alloy steels.

"6.1.12 MIL-312-15 and 16. These electrodes are intended for the joining of the 'super-alloys' to low-alloy (chromium-molybdenum, and chromium-nickel-molybdenum) steels.

"6.1.13 MIL-316-15 and 16. These electrodes are intended for the joining of corrosion-resistant alloys of similar designations. The molybdenum content imparts increased resistance to corrosive effect of many organic acids, brine, sulfurous and sulfuric acid.

"6.1.14 MIL-316L-15 and 16. These electrodes are intended for the welding of type 316L corrosion-resisting steel and under conditions similar to those under which type MIL-318 could be used.

"6.1.15 MIL-317-15 and 16. These electrodes are intended for welding of alloys of similar composition for increased corrosion resistance to sulfuric and sulfurous acid and their salts.

"6.1.16 MIL-318-15 and 16. These electrodes are intended for the welding of type 318 corrosion-resisting steel. Generally, uses are similar to those of type MIL-316 electrode, except that, where elevated temperatures and corrosive conditions favor intergranular corrosion and type 318 alloy is used, choice of stabilized type MIL-318 electrode is indicated.

"6.1.17 MIL-330-15 and 16. These electrodes are used in repair of alloy castings and welding of cast and wrought alloys of similar compositions which are generally used for heat-resisting properties of high temperatures.

"6.1.18 MIL-347-15 and 16. These electrodes are intended for the general welding of 18-8 corrosion-resisting steels stabilized by columbium, which are used where conditions of welding or service make unstabilized base metals susceptible to carbide precipitation, with consequent intergranular corrosion. Use of these electrodes will not prevent intergranular corrosion alongside a weld in unstabilized 18-8 chromium-nickel base metals.

"6.1.19 MIL-347Co-15 and 16. These electrodes are intended for the welding applications of 6.1.18 requiring a limited cobalt content.

"6.1.20 MIL-347HC-15 and 16. These electrodes are intended for applications where low magnetic permeability or heat resistance is of prime importance and where corrosion is not a factor.

"6.1.21 MIL-349-15 and 16. These electrodes are intended for the joining of the 'super-alloys' for applications requiring high strength at elevated temperatures.

"6.1.22 MIL-16.8.2-15 and 16. These electrodes are intended for use in welding 316, 317, and 347 alloys for corrosion resisting and high temperature service in as-welded or stress-relieved condition."

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Page 10

6.3: Delete and substitute:

"6.3 Inspection after delivery. Post delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee. When receipt inspection testing types MIL-308, MIL-308Co, MIL-308L, MIL-308CoL, MIL-308HC, MIL-309, MIL-309Co, MIL-309L, MIL-309CoL, MIL-309Cb, MIL-347 and MIL-347Co, the consignee or user activity may consider the following delta ferrite test result limits acceptable:

(a) A maximum of 12 percent delta ferrite.
(b) An average minimum level of 3 percent delta ferrite with no individual inspection level of less than 2 percent delta ferrite.

Vendor test results shall meet the specified 4 to 9 percent delta ferrite content."

Page 17

Figure 5: Delete and substitute the attached figure 5.

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CERTIFICATION OF QUALITY CONFORMANCE TESTS[1/][2/]

Manufacturer or Distributor_____ Customer's Name_____

Address_____ Customer's Order No._____

Date_____ Core Wire Heat No._____

Specification MIL-_____ Lot Identification MIL-E-0022200E Para._____

Type MIL- _____

Diameter & Length _____ Ferrite Content

Inspection Level _____ Percent ferrite (average)

Lot No._____

Chemical Analysis

Carbon_____

Manganese_____

Phosphorus_____

Sulfur_____

Silicon_____

Chromium_____

Nickel_____

Molybdenum_____

Columbium_____

Tungsten_____

Cobalt_____

Groove Weld Test

Chemistry was taken from: ☐ Chem Pad Test No. 8a Chem Pad 3 6

☐ Groove Weld Amperage _____ _____ ____ ____

X-Ray Results_____

Concentricity (%)_____ Grinding during 8a Test Plate Preparation[3/] Operator Error (Layer Nos.)

Covering Moisture (QPL) (%)_____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____

[1/] Items not applicable to the type electrode involved or exempted by the procurement document shall be marked N/A.

[2/] This form is applicable to the quality conformance tests required for each lot and a copy shall be forwarded with each shipment.

[3/] See 3.5.7 and 3.9 of MIL-E-0022200E.

Figure 5

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